# Exercise 4

**AIM :**

To Write a C program to input marks of five subjects . Calculate percentage and grade according to following:
Percentage >= 90% : Grade A
Percentage >= 80% : Grade B
Percentage >= 70% : Grade C
Percentage >= 60% : Grade D
Percentage >= 40% : Grade E
Percentage < 40% : Grade F

**ALGORITHM :**

Step 1: PROGRAM INITIALIZATION

- Define necessary functions for modular programming
- Declare main variables: marks array, total, percentage, grade

Step 2: INPUT PHASE

- Display prompt for entering marks
- Use a for loop to collect marks for 5 subjects
- Validate each input (0-100 range)
- If invalid input detected, decrement counter to re-enter for same subject

Step 3: CALCULATION PHASE

- Total Calculation: Sum all marks using calculateTotal() function
- Percentage Calculation: Compute using calculatePercentage(): (Total/500)×100
- Grade Determination: Use determineGrade() with grade boundaries

Step 4: OUTPUT PHASE

- Display formatted result using displayResult() function
- Show individual subject marks
- Display total marks, percentage, and final grade
- Present results in a clear, tabular format

Step 5: PROGRAM TERMINATION

- Return 0 indicating successful execution

**OUTPUT :**

```
D:\C programming\DHAVAMANI_EX4>gcc main.c -o percent.exe

D:\C programming\DHAVAMANI_EX4>percent.exe
Enter marks for 5 subjects (each out of 100):
Subject 1: 99
Subject 2: 98
Subject 3: 97
Subject 4: 96
Subject 5: 95

======== RESULT ========
Subject 1: 99.00
Subject 2: 98.00
Subject 3: 97.00
Subject 4: 96.00
Subject 5: 95.00
-------------------------
Total Marks: 485.00/500
Percentage: 97.00%
Grade: A
=========================
```